Proceedings of the 17th IEEE International
Conference on Nano/Micro Engineered and Molecular Systems
April 14-17, 2022

2022 IEEE 17th International Conference on Nano/Micro Engineered and Molecular Systems (NEMS) | 978-1-6654-8301-8/22/$31.00 ©2022 IEEE | DOI: 10.1109/NEMS54180.2022.9791144

Detection of Molecules Based on Enhanced Backscattering Effect in Microsphere Lens

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***Abstract*—Owning to the enhanced backscattering effect in the photonic nanojet, detection approach based on measuring the backscattering signal allows for the detection of micro- and nanoscale objects with single-object resolution. However, this approach fails in the detection of molecules, since the weak light-matter interaction between photonic nanojet and molecules is always insufficient to excite detectable backscattering signal. To address this issue, we employ signal amplification via ternary assembly of lens-aptamer-nanoparticle complex to enhance the backscattering signal triggered by molecules. We show that this approach reveals pronounced enhancement on backscattering signals and therefore allows successive detection of molecules in the solution. This approach provides a step forward in facilitating photonic nanojet towards practical application in the field of highly sensitive detection of bio/chemical molecules.**

I. Introduction

Microsphere lenses refer to transparent spherical dielectric microspheres with a diameter of several micrometers to tens of micrometers. When illuminated by light waves, different from the traditional lenses of larger sizes, the micron-level microsphere lens can generate a beam of light waves with extremely high intensity and narrow waist in the shadow part of the microsphere, which is called photonic nanojet (PNJ) [1-3]. Nanometric object that enters into the PNJ area can strongly interact with the PNJ and excite a greatly enhanced backscattering signal with a gain coefficient of up to 6 orders of magnitude (million times) referred as enhanced backscattering (EBS) effect [4-6]. The EBS effect results from the synergistic effect between the high-intensity backscattered light excited by the object in the PNJ and the efficient optical collection of the backscattered light by the microsphere lens. This effect has enabled the PNJ as a useful tool to optically detect the micro- or nanoscale objects and it has been successfully utilized to detect individual nanoparticles, cells, bacteria, etc. [7-10] However, the true potential of this approach is still behind expectations with respect to the ability for molecular detection. Due to the ultra-small molecular effective radius in solution, the weak light-matter interaction between the PNJ and molecules is insufficient to excite detectable backscattering signals, thus no EBS effect can be recorded. The above described limitation is simply an intrinsic property in the EBS effect, greatly hindering the possible applications of the PNJ in general for biosensing applications. In this contribution, we propose an approach that can strongly amplify the intensity of the backscattering signal triggered by molecules and therefore allows the detection of molecules via EBS effect in the microsphere lens.

II. Methods

1. *Fabrication of microwell array*

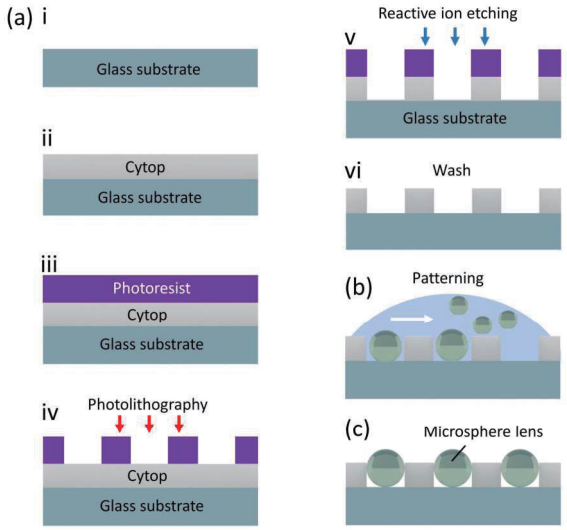

Figure 1. (a) Schematic illustration of the fabrication process of microwell array. i and ii: Cytop is coated on the top of the glass substrate. iii: Photoresist is spin-coated on the Cytop layer. iv: The wafer is treated with standard photolithography. v: The photoresist layer and Cytop layer are etched by reactive ion etching. vi: Photoresist layer is removed, leaving the microwell array. (b) Melamine microspheres are patterned on the microwell array. (c) The formed microsphere lens array.

The microwell array was fabricated on the glass substrate, as reported in previous study [11]. Briefly, transparent float glass wafer of 4 inch was used as the substrate. Oxygen plasma was used to clean the substrate and to promote its affinity to the upcoming spin coating layer. Cytop-809M was used as the spin coating layer and a thickness of around 3μm was achieved. Then the Cytop layer was cured by placing on a hot plate. A gradient temperature was increased from room temperature to 180 ℃ to cure the Cytop layer. When the Cytop layer was cured, photoresist AZ4620 (8 μm thickness) was spin-coated on the Cytop layer. Afterwards, the glass substrate was treated with

Research supported by the National Natural Science Foundation of China (NSFC, No. 62175252, 62074155), Guangdong Basic and Applied Basic Research Foundation (2019A1515011242), Shenzhen Science and Technology Innovations Committee (JCYJ20210324101405016, JSGG20210713091807026).

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standard photolithography and then washed to form the pattern on the photoresist. O_2 reactive ion etching (RIE) was used to etch on the Cytop layer. After etching, the photoresist layer was then removed, leaving the microwell array pattern on the Cytop layer, Figure 1(a). The diameter of the microwell is ~3.5 μm and they were arranged in quadrilateral pattern.

2. *Patterning of the melamine microspheres on the microwell array*

Template-assisted self-assembly (TASA) was used to pattern the melamine microspheres on the microwell array, Figure 1(b). The detailed approach was also reported in previous study [11]. This could generate microsphere lens array with well-controlled spatial layouts on surfaces patterned microwell array. The carboxyl-functionalized melamine microspheres with 3 μm in diameter were used to be patterned on the microwell array. Briefly, melamine microspheres in buffer solution with pH = 1.0 was used as the patterning solution. The patterning process was conducted on a homemade self-assembly setup. An ordered array of melamine microsphere lens can be obtained after the pattering, Figure 1(c).

3. *Surface modification of the gold nanoparticles and the melamine microspheres*

Adenosine triphosphate (ATP) split aptamer was used: fragment (1) 5'-HS-SH-(CH2)6-AAAAAAAAAA TGC GGAGGAAGGT-3', and fragment (2) 5'-NH2-(CH2)6-AAAAAAAAAAA CCTGGGGGAGTAT-3'. Commercial gold nanoparticles with diameter of 60nm was used. Fragment (1) of ATP split aptamer with thiol terminal was modified on the gold nanoparticle (AuNP) via salt-aging method. Fragment (2) of ATP split aptamer with amino terminal was modified on the carboxyl-functionalized melamine microspheres via click chemistry to form covalent coupling.

4. *Simulation of photonic nanojets and the backscattering signal on the presence of gold nanoparticle*

Simulation based on finite element method (FEM) was conducted to study the generation of the photonic nanojets and the backscattering signal on the presence of gold nanoparticle. The refractive index of the surrounding medium was set at 1.33. The AuNP was placed on the topside of the microsphere lens. The backscattering signal with and without AuNP was calculated.

5. *Optical measurement under the light microscope*

The sample was characterized under the inverted light microscope equipped with visible light source. Figure 2. The backscattering signal was collected by a CMOS camera. The objective was focused on the focal plane of the microsphere lens.

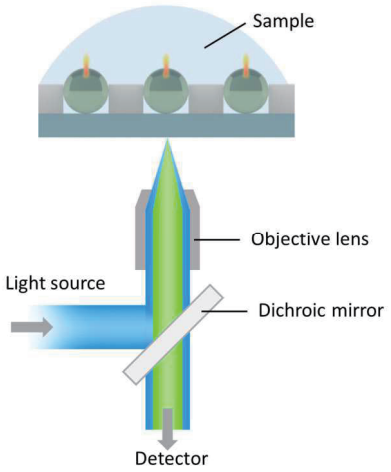

Figure 2. Schematic illustration of the optical setup used in the experiments.

III. Results and Discussion

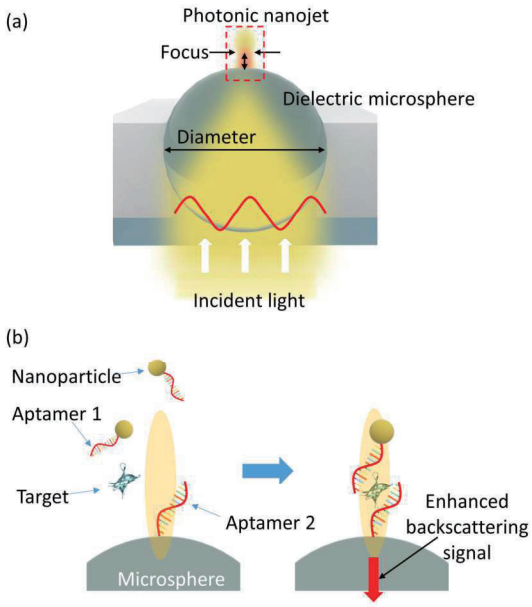

Figure 3. Schematic illustration of the molecular detection mechanism based on the enhanced backscattering effect in photonic nanojet. (a) Generation of the photonic nanojet by the melamine microsphere. (b) Formation of the ternary complex "lens-aptamer-nanoparticle" on the presence of target (ATP) molecules.

As a model system, we chose adenosine triphosphate (ATP) as the target molecule. The detection mechanism based on EBS effect is depicted in Figure 3. Briefly, melamine microsphere with diameter of 3um is illuminated beneath to generate PNJ on the top, Figure 3(a). Two fragments of the ATP split aptamer are separately modified on the gold nanoparticles (AuNPs) and on the surface of the melamine microspheres, respectively. The presence of ATP in the solution would trigger a specifically formation of ternary complex (lens-aptamer-nanoparticle) assembled by these two fragments as well as the AuNPs, Figure 3(b). This process simultaneously introduces the AuNPs into the jet area and induces the EBS effect in the PNJ. The as generated backscattering signal can be recorded from far field through conventional optical microscope, indicating a molecular detection event and achieving the detection of molecules.

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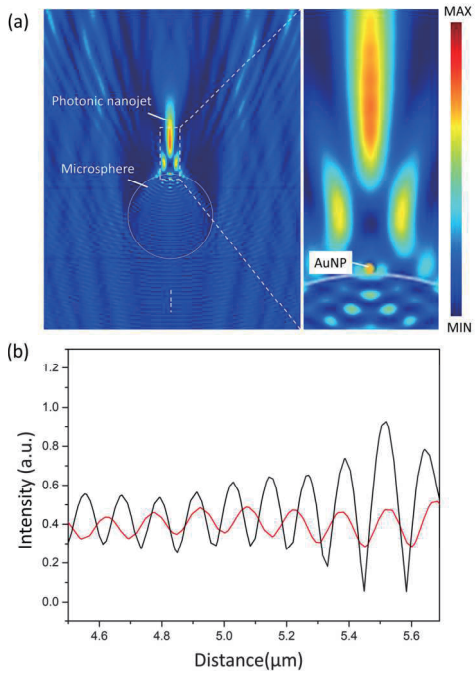

Figure 4. (a) Finite-difference time-domain simulation of the AuNP induced EBS effect and (b) the enhanced backscattering signal (black line) compare with the signal of simply induced by AuNP (red line).

Simulation based on finite element method (FEM) is performed to show the enhanced backscattering signal induced by a single AuNP on the surface of the melamine microsphere, Figure 4(a). The AuNP is set on the topside of the melamine microsphere. The backscattering signal of with and without the AuNP is shown in Figure 4(b). It is shown that, on the presence of the AuNP, the backscattering signal is obviously increased. Up to 2-fold enhancement on the intensity of the signal can be obtained. The simulation confirms the entrance of the AuNP into the PNJ can significantly disturbs the backscattering signal of the system.

To observe numerous detection events simultaneously, array of assembled melamine microsphere was utilized in the experiment. The melamine microspheres were assembled in the microwell via template-assisted self-assembly (TASA) approach and their scanning electron microscope (SEM) image is shown in Figure 5. It is observed that the microsphere lenses are all loaded into the microwells, forming a quadrilateral pattern. The fabricated ordered array of melamine microsphere lens provides the possibility of high-throughput observation.

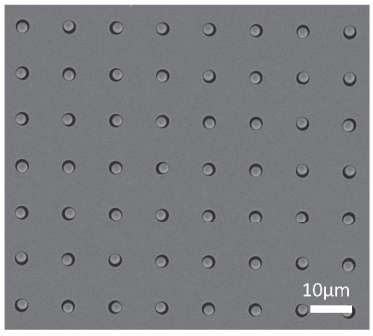

Figure 5. Scanning electron microscope (SEM) image of the array of assembled melamine microspheres.

The fabricated ordered array of melamine microsphere lens were modified with ATP split aptamer as described in the methods section. Afterwards, the array was immersed in the detection buffer and was observed under the inverted light microscope equipped with visible light source and CMOS camera. The backscattering signals of the melamine microsphere were collected and their intensity can be obtained. On the absence of ATP in the solution, fleeting backscattering signals are observed, caused by free AuNPs that randomly pass the PNJ area due to their Brownian motion in the solution. However, on the presence of ATP, sustained backscattering signals are observed, which indicates the ATP triggered formation of stable ternary complex in the PNJ area, Figure 6(a). The intensity of the backscattering signal on the presence of the AuNP is increased about 12% in Figure 6(b), which can be clearly observed in the microscope.

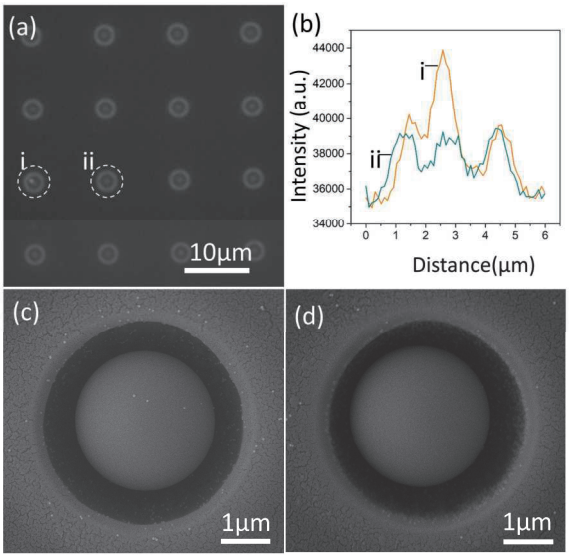

Figure 6. (a) Intensities of the backscattering signals of the arrayed microsphere lens under white-light illumination: (i) EBS effect generated sustained enhanced intensity and (ii) normal intensity with no EBS effect. (b) The corresponding intensity profiles of (i) and (ii), showing a ~1.12 folds of intensity difference. SEM images of the melamine spheres (c) exhibiting enhanced backscattering intensity and (d) no enhanced backscattering intensity. The image of (c) confirms the enhanced backscattering signal is excited by the AuNPs on the melamine microspheres.

The SEM images further confirm the sustained signals are excited by the AuNPs on the jet area of the melamine microspheres, Figure 6 (c) and (d). For the microsphere lens without sustained backscattering signals under the light microscope, no AuNPs is observed on the microsphere lens. While for the microsphere lens with sustained backscattering signals, AuNPs are observed on the topside. It should be noted that, the intensity of the backscattering signal increases if more AuNPs are attached on the microsphere lens. Meanwhile, the concentration of the ATP molecules in the detection buffer also influence the number of detection events in the lens array. Further experiments were conducted to study the intensity of the backscattering signal and the number of the detection events in order to quantify the concentration of the target molecules.

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IV. Conclusion

In summary, we have developed a molecule detection approach based on enhanced backscattering effect in microsphere lens. The microsphere lenses are ordered patterned on the transparent substrate. The formation of ternary complex "lens-aptamer-nanoparticle" on the presence of target molecules induces sustained backscattering signals, which can be observed under the light microscope, achieving the detection of molecules. A pronounced enhancement on backscattering signals is observed, demonstrating its feasibility for facilitating photonic nanojet towards practical application. This approach is promising to be used for ultra-sensitive detection of bio/chemical molecules.

Acknowledgment

The authors acknowledge financial support from the National Natural Science Foundation of China (NSFC, No. 62175252, 62074155), Guangdong Basic and Applied Basic Research Foundation (2019A1515011242), Shenzhen Science and Technology Innovations Committee (JCYJ20210324101405016, JSGG20210713091807026).

References

[1] A. Heifetz, S. C. Kong, A. V. Sahakian, A. Taflove, and V. Backman, "Photonic nanojets," *J Comput Theor Nanosci,* vol. 6, pp. 1979-1992, 2009.

[2] S. Lecler, S. Perrin, A. Leong-Hoi, and P. Montgomery, "Photonic jet lens," *Sci Rep,* vol. 9, pp. 4725, 2019.

[3] B. S. Luk'yanchuk, R. Paniagua-Domínguez, I. Minin, O. Minin, and Z. Wang, "Refractive index less than two: photonic nanojets yesterday, today and tomorrow," *Opt Mat Express,* vol. 7, pp. 1820, 2017.

[4] Z. Chen, A. Taflove, and V. Backman, "Photonic nanojet enhancement of backscattering of light by nanoparticles: a potential novel visible-light ultramicroscopy technique," *Opt Express,* vol. 12, pp. 1214-20, 2004.

[5] X. Li, Z. Chen, A. Taflove, and V. Backman, "Optical analysis of nanoparticles via enhanced backscattering facilitated by 3-D photonic nanojets," *Opt Express,* vol. 13, pp. 526-33, 2005.

[6] Z. Chen, A. Taflove, X. Li, and V. Backman, "Superenhanced backscattering of light by nanoparticles," *Opt Lett,* vol. 31, pp. 196-8, 2006.

[7] H. Yang, M. Cornaglia, and M. A. Gijs, "Photonic nanojet array for fast detection of single nanoparticles in a flow," *Nano Lett,* vol. 15, pp. 1730-5, 2015.

[8] Y. Li, H. Xin, X. Liu, Y. Zhang, H. Lei, and B. Li, "Trapping and detection of nanoparticles and cells using a parallel photonic nanojet array," *ACS Nano,* vol. 10, pp. 5800-8, 2016.

[9] Y. C. Li, H. B. Xin, H. X. Lei, L. L. Liu, Y. Z. Li, Y. Zhang*, et al.*, "Manipulation and detection of single nanoparticles and biomolecules by a photonic nanojet," *Light Sci Appl,* vol. 5, pp. e16176, 2016.

[10] P. Zhang, B. Yan, G. Gu, Z. Yu, X. Chen, Z. Wang*, et al.*, "Localized photonic nanojet based sensing platform for highly efficient signal amplification and quantitative biosensing," *Sensors and Actuators B: Chemical,* vol. 357, pp. 131401, 2022.

[11] P. Zhang, X. Chen, and H. Yang, "Large-scale fabrication of photonic nanojet array via template-assisted self-assembly," *Micromachines,* vol. 11, pp. 473, 2020.

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